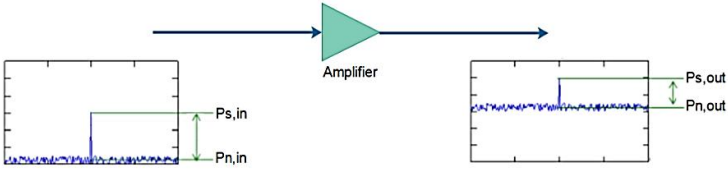
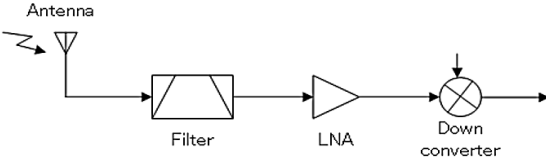
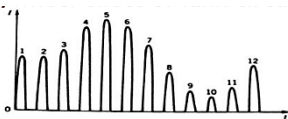
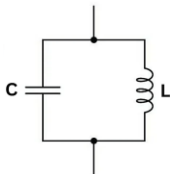
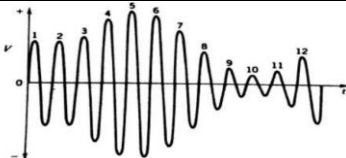
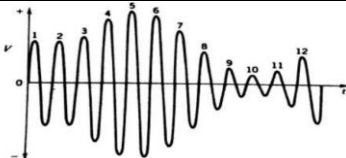
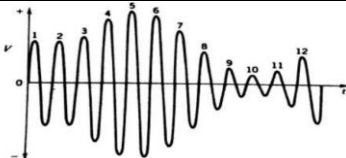
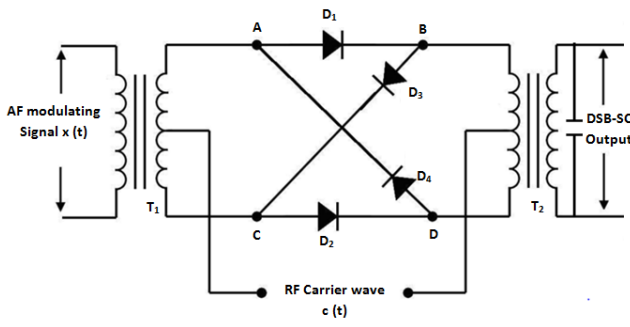
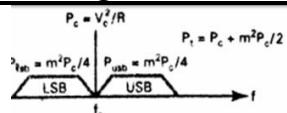
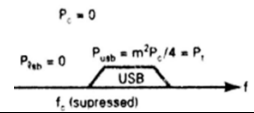
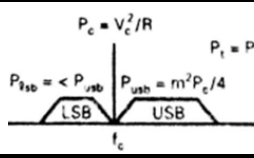
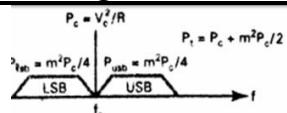
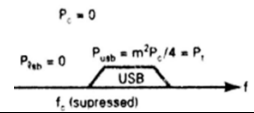
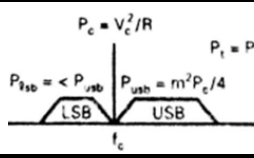
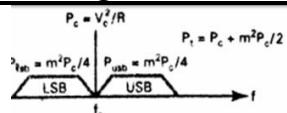
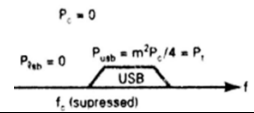
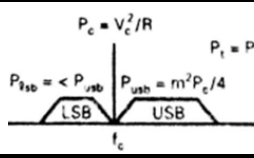


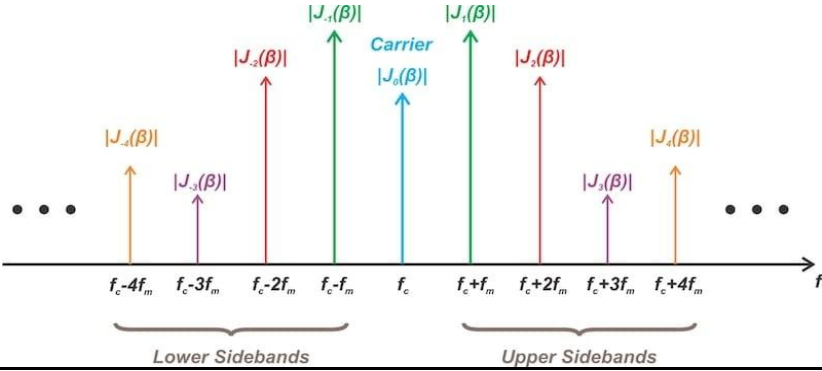
Semester: January 2026 – April 2026		
Maximum Marks: 30	Examination: In-Semester Examination	Duration: 1hr.15 Min
Programme code: 03 Programme: B.Tech Electronics and Telecommunication Engineering	Class: SY	Semester: IV (SVU 2023)
Institute/School/ Department: K. J. Somaiya School of Engineering	Name of the department: EXTC	
Course Code: 216U03C403	Name of the Course: Communication Systems	
Note: All questions are of TWO marks (Model Answer Key)		

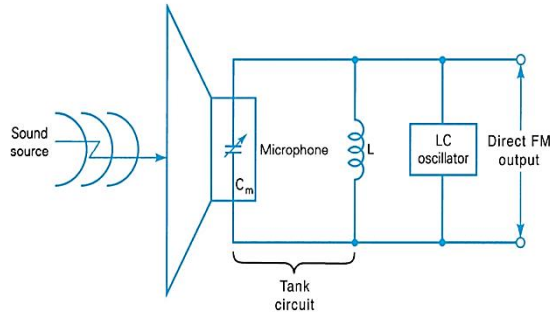
Q.1	Attempt any <u>FIVE</u> of the following		5 × 2 = 10
Q.1 (a)	Draw neat labeled block diagram of electronics communication system.		
	Only diagram	1 mark	
	Labeling of all blocks with example	1 mark	
Q.1 (b)	Calculate the height of antenna for frequency of 10 KHz and 100 MHz. Which value of antenna height is preferred for communication system? Why?		
	When $f = 10 \text{ KHz}$, $\lambda = \frac{c}{f} = 30 \text{ km}$ and height of antenna is $\frac{\lambda}{2} = 15 \text{ km}$	0.5 marks	
	When $f = 100 \text{ MHz}$, $\lambda = \frac{c}{f} = 3 \text{ m}$ and height of antenna is $\frac{\lambda}{2} = 1.5 \text{ m}$	0.5 marks	
	When $f = 10 \text{ KHz}$ the height of antenna is 15 km which is impractically large and expensive to construct. When $f = 100 \text{ MHz}$, the height of antenna is 1.5 m which is small and practical and easier to install. Therefore 1.5 m antenna height is preferred for communication system.		1 mark
Q.1 (c)	Write applications of following bands in electromagnetic spectrum (i) Radio wave (ii) Infrared ray (iii) Ultraviolet rays (iv) Gamma rays		
	Radio waves	Radio and TV broadcasting, mobile and wireless communication	0.5 marks
	Infrared rays	TV remote control, thermal imaging, heat therapy	0.5 marks
	Ultraviolet rays	Sterilization of medical instruments, water purification, forensic investigation	0.5 marks
	Gamma rays	Cancer treatment, food preservation, nuclear research and medical imaging	0.5 marks
Q.1 (d)	Explain the term “uncorrelated noise”. Give examples of it.		
	Definition of uncorrelated noise	1 mark	
	Shot noise, thermal noise, external noise, solar noise, cosmic noise etc.	1 mark	
Q.1 (e)	Write your interpretation from the following figure.		
			2 marks
	Figure depicts the system with input signal to noise power ratio and output signal to noise power ratio. At input signal to noise ratio is high. When signal passes through the system (amplifier/filter etc.) it adds noise to it and hence at output signal to noise ratio decreases. From this we can calculate noise factor and noise figure of the system, which are		

	important figure of merit parameters for any communication systems							
Q.1 (f)	For the communication receiver shown below calculate over all Noise Figure, if noise figure of each stage is 3 dB and power gain is 10 dB.  <table><tr><td>Given is a 3-stage receiver circuit Noise figure (F) = 3 dB = 2 (linear scale) Gain (G) = 10 dB = 10 (linear scale)</td><td>1 mark</td></tr><tr><td>$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} = 2.11$NF(total)=10 log (2.11) = 3.24 dB</td><td>1 mark</td></tr></table>	Given is a 3-stage receiver circuit Noise figure (F) = 3 dB = 2 (linear scale) Gain (G) = 10 dB = 10 (linear scale)	1 mark	$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} = 2.11$ NF(total)=10 log (2.11) = 3.24 dB	1 mark			
Given is a 3-stage receiver circuit Noise figure (F) = 3 dB = 2 (linear scale) Gain (G) = 10 dB = 10 (linear scale)	1 mark							
$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} = 2.11$ NF(total)=10 log (2.11) = 3.24 dB	1 mark							
Q.2	Attempt any <u>SIX</u> of the following	6 × 2 = 12						
Q.2 (a)	Define the term coherent and non-coherent detectors. <table><tr><td>Coherent detector: The detectors which requires carrier (reference) signal for demodulation are called coherent detectors</td><td>1 mark</td></tr><tr><td>Non-coherent detectors: The detectors which do not require carrier (reference signal) for demodulation are called non-coherent detectors.</td><td>1 mark</td></tr></table>	Coherent detector: The detectors which requires carrier (reference) signal for demodulation are called coherent detectors	1 mark	Non-coherent detectors: The detectors which do not require carrier (reference signal) for demodulation are called non-coherent detectors.	1 mark			
Coherent detector: The detectors which requires carrier (reference) signal for demodulation are called coherent detectors	1 mark							
Non-coherent detectors: The detectors which do not require carrier (reference signal) for demodulation are called non-coherent detectors.	1 mark							
Q.2 (b)	Define Amplitude Modulation. Draw output waveform of the following circuit for the given input.  Input  Output = ? <table><tr><td>Definition of amplitude modulation</td><td>1 mark</td></tr><tr><td></td><td>1 mark</td></tr><tr><td>Output of the circuit is:</td><td></td></tr></table>	Definition of amplitude modulation	1 mark		1 mark	Output of the circuit is:		
Definition of amplitude modulation	1 mark							
	1 mark							
Output of the circuit is:								
Q.2 (c)	What is the difference between conventional AM and ISB? Which one is more beneficial and why? <table><tr><td> In conventional AM there are two identical sidebands (USB and LSB) compared with in ISB there are two independent sidebands. AM has same information in both sideband and ISB has different information, Bandwidth of AM is 2fm and ISB is also 2fm but carries double information </td><td>1 mark</td></tr><tr><td>ISB is more beneficial because it allows transmission of two different signals simultaneously using upper and lower sidebands. Better bandwidth utilization (more information in same bandwidth). Improved efficiency compared with</td><td>1 mark</td></tr></table>	In conventional AM there are two identical sidebands (USB and LSB) compared with in ISB there are two independent sidebands. AM has same information in both sideband and ISB has different information, Bandwidth of AM is 2fm and ISB is also 2fm but carries double information	1 mark	ISB is more beneficial because it allows transmission of two different signals simultaneously using upper and lower sidebands. Better bandwidth utilization (more information in same bandwidth). Improved efficiency compared with	1 mark			
In conventional AM there are two identical sidebands (USB and LSB) compared with in ISB there are two independent sidebands. AM has same information in both sideband and ISB has different information, Bandwidth of AM is 2fm and ISB is also 2fm but carries double information	1 mark							
ISB is more beneficial because it allows transmission of two different signals simultaneously using upper and lower sidebands. Better bandwidth utilization (more information in same bandwidth). Improved efficiency compared with	1 mark							

	conventional AM where one sideband is redundant																			
Q.2 (d)	Draw neat labeled circuit diagram of Ring modulator for DSBSC generation. Neat labeled circuit diagram: 	2 marks																		
Q.2 (e)	The equation of AM wave is given by: $s(t) = 20[1 + 0.8 \cos(2\pi \times 10^3 t)] \cos(4\pi \times 10^5 t)$ Compute carrier power, total sideband power. <table><tr><td>Carrier amplitude $V_c = 20$ and therefore $P_c = \frac{V_c^2}{2} = \frac{400}{2} = 200 \text{ watts}$</td><td>1 mark</td></tr><tr><td>Total sideband power $\frac{m^2}{2} \times P_c = 0.32 \times 200 = 64 \text{ Watts}$</td><td>1 mark</td></tr></table>		Carrier amplitude $V_c = 20$ and therefore $P_c = \frac{V_c^2}{2} = \frac{400}{2} = 200 \text{ watts}$	1 mark	Total sideband power $\frac{m^2}{2} \times P_c = 0.32 \times 200 = 64 \text{ Watts}$	1 mark														
Carrier amplitude $V_c = 20$ and therefore $P_c = \frac{V_c^2}{2} = \frac{400}{2} = 200 \text{ watts}$	1 mark																			
Total sideband power $\frac{m^2}{2} \times P_c = 0.32 \times 200 = 64 \text{ Watts}$	1 mark																			
Q.2 (f)	State whether following statements are True or False and justify: (i) The bandwidth of VSB is slightly more than that of SSB. (ii) SSB has lower power efficiency compared to DSBFC. <table><tr><td>(i) True (ii) False</td><td>1 mark each</td></tr></table>		(i) True (ii) False	1 mark each																
(i) True (ii) False	1 mark each																			
Q.2 (g)	Match the following columns: <table><tr><td>1.</td><td></td><td>A.</td><td>SSBFC</td></tr><tr><td>2.</td><td>VSB</td><td>B.</td><td></td></tr><tr><td>3.</td><td></td><td>C.</td><td>DSBFC</td></tr><tr><td>4.</td><td>SSBSC</td><td>D.</td><td></td></tr></table> <table><tr><td>1-C, 2-D, 3-A, 4-B</td><td>½ mark each</td></tr></table>		1.		A.	SSBFC	2.	VSB	B.		3.		C.	DSBFC	4.	SSBSC	D.		1-C, 2-D, 3-A, 4-B	½ mark each
1.		A.	SSBFC																	
2.	VSB	B.																		
3.		C.	DSBFC																	
4.	SSBSC	D.																		
1-C, 2-D, 3-A, 4-B	½ mark each																			
Q.2 (h)	Name the following SSB modulator and draw the frequency spectrum at point A, B and C for the give input frequency spectrum.																			

	B = 5 kHz 0 5k Input Modulating signal input Amp Balanced modulator BPF sum. Summer LF carrier osc. 100 kHz Carrier pilot adjust Buffer amp B C	
	Name of figure: SSB modulator using Filter method ½ mark DSBSC B = 10 kHz 95k 100k 105k Waveform at point A → SSBSC B = 5 kHz 100k 105k Waveform at point B → SSBRC B = 5 kHz 100k 105k Waveform at point C →	½ mark ½ mark ½ mark ½ mark
Q.2 (i)	For the AM waveform shown below, calculate modulation index. Amplitude (V) </	

	$\omega_i = \text{_____} + \Delta f$	
	$\Delta f = \text{_____} \times v_m(t)$	
	$\theta = \omega_c t + \text{_____} \times \sin \omega_m t$	
	$\text{_____} = \frac{d\theta}{dt}$	
	$\omega_i = \omega_c + \Delta f$	½ mark
	$\Delta f = k_f \times v_m(t)$	½ mark
	$\theta = \omega_c t + m_f \times \sin \omega_m t$	½ mark
	$\omega_i = \frac{d\theta}{dt}$	½ mark
Q.3 (c)	Define the term instantaneous phase and instantaneous phase deviation.	
	Instantaneous phase: It is a precise phase of the carrier at a given instant of time.	1 mark
	Instantaneous phase deviation: It is the instantaneous change in the phase of the carrier at a given instant of time with reference to phase.	1 mark
Q.3 (d)	Draw neat labeled frequency spectrum of FM wave.	
	Neat labeled diagram: 	2 marks
Q.3 (e)	Find the instantaneous frequency in Hz for the given signal below: $10 \cos(200\pi t + \frac{\pi}{3})$	
	Comparing with $f_m(t) = v_c \cos \theta_i(t)$ $\omega_i = \frac{d\theta_i(t)}{dt}$ $\theta_i(t) = 200\pi t + \frac{\pi}{3}$ $\omega_i = \frac{d}{dt} [200\pi t + \frac{\pi}{3}]$ $\omega_i = 200\pi \text{ rad/sec}$	Calculation of ω_i 1 mark and calculation of f_i 1 mark $f_i = \frac{\omega_i}{2\pi}$ $f_i = \frac{200\pi}{2\pi}$ $f_i = 100 \text{ Hz}$
Q.3 (f)	Name the circuit given below, which elements of the circuit decide carrier frequency and which element decides frequency deviation.	



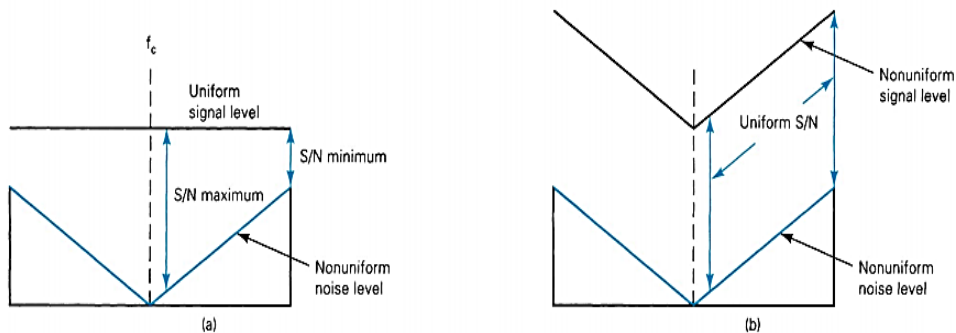
The circuit given below is direct FM modulator circuit

1 mark

The carrier rest frequency is decided by LC oscillator (when NO modulating signal is applied). Frequency deviations are produced by capacitor C_m and inductor L connected in parallel with LC oscillator. The values of these components are proportional to change in modulating signal (sound source) in given figure.

1 mark

Explain following figure (a) and (b) with reference to FM:



Q.3 (g)

Figure (a) shows uniform signal level at low and high frequency which produces high SNR at low frequency but when signal frequency increases SNR reduces (at high frequency) which will produce noise in high frequency

1 mark

Figure (b) shown artificial boosting of high frequency signal through pre-emphasis network at transmitter and uniform SNR is maintained for all frequencies.

1 mark